

PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENTS
CHAMWINO DISTRICT COUNCIL
FORM FOUR JOINT EXAMINATION

031/1

PHYSICS 1

TIME: 3:00 Hours

wazaelimu.com

MARCH 2026

INSTRUCTIONS

1. This paper consists of sections **A, B** and **C** with a total of eleven (11) questions.
2. Answer **ALL** questions in section **A** and **B** and **two (2)** questions from section **C**
3. Cellular phones and unauthorized materials are not allowed in the examination room
4. Non- programmable calculator may be used
5. Write your examination number on every page of your answer **sheets/booklet(s)**
6. Whenever necessary, the following constants may be used;
 - Acceleration due to gravity, $g=10\text{m/s}$
 - Pie, $\pi = 22/7$
 - Density of water $=1.0\text{g/cm}^3$ or 1000kg/m^3
 - Density of sea water 3 1.3g / cm^3 or 3 1300kg /m^3

SECTION A (16 Marks)

Answer **all** questions in this section

1. For the item (i) - (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet(s)/booklet provided;
 - (i) When using a measuring cylinder one precaution to be taken is to
 - A. Check for zero error
 - B. Look at the meniscus from below the level of the water
 - C. Obtain more readings by looking from more than one direction
 - D. Positioning the eye in line with base of the meniscus
 - E. Make sure that the measuring cylinder is clean
 - (ii) Which of the following sources of energy are non renewable?
 - A. Water, wind, wood and natural gas
 - B. Fossils, sun, oil, and nuclear
 - C. Natural gas, water nuclear, and wood
 - D. Wind, sun, fossils, and oil
 - E. Oil, and all natural gases

(iii) A material which allows some light to pass through it but one cannot see through it is said to be;

- A. Transparent B. Translucent C. Luminous D. Opaque E. Colourless

(iv) Which phenomenon causes the dispersion of white light into a spectrum by prism?

- A. Diffraction B. Absorption C. Interference D. Refraction E. Reflection

(v) Two forces of 5N and 8N are acting at the same point and are inclined at an angle of $0^\circ 45'$ to each other. What will be their resultant force?

- A. 11.2N B. 12N C. 22.4N D. 1.2N E. 1.12N

(vi) A bus carrying a very big load on its top carrier can easily overturn because;

- A. It cannot run fast D. It is more stable
B. Its equilibrium is stable E. Its centre of gravity is high
C. Its centre of gravity is low

(vii) Which of the following is not a property of magnetic lines of force due to a bar magnet?

- A. They have a direction from North to South pole outside the magnet
B. They do not exist inside the magnet
C. They have direction from South pole to North pole inside the magnet
D. They tend to be closer inside but wider apart outside the magnet
E. They form complete loop

(viii) A certain wave has a period of 0.2 seconds and a wavelength of 60cm. What is the velocity of wave in cm/s?

- A. 30 B. 0.30 C. 12 D. 3 E. 300

(ix) A diode is an electronic component used to;

- A. Rectifying oscillating current
B. Amplifying signal
C. Rectifying oscillating current and detect electrical signals
D. Amplify current and detect electrical signals
E. Rectify the voltage

(x) A rectangular wooden block of density 0.8g/cm^3 has dimensions $0.5 \times 0.8 \times 6\text{m}$. What maximum pressure will it exert on a ground in N/M^2 ?

- A. 48000 B. 4000 C. 2.4 D. 19200 E. 400

2. Match the descriptions of the terms used in simple machines in List A with their corresponding concept used in simple machines in List B.

LIST A	LIST B
(i) Is useful in drawing water from a borehole (ii) Applies the principle of first-class lever (iii) Its velocity ratio increases by increasing the length of the ramp (iv) The effort is between the load and fulcrum (v) Operates based on the principle of transmission of pressure in fluids (vi) Carries objects from one point to another horizontally	A. Wheelbarrow B. Wheel and axle C. Hydraulic press D. Tong E. Claw hammer F. Inclined plane G. Pulley H. Windmill I. Screw jack

SECTION B (54 MARKS)

Answer **all** questions in this section

3. (a) A convex mirror is often used as the wing mirror of a car instead of plane mirror. Justify this statement giving one point. **(3 marks)**
- (b) A camera is used to take a close-up picture of an object 3 cm tall. If the object is positioned 24 cm in front of the lens and a focused image is formed on the film 12 cm behind the lens, determine the focal length of the camera lens and the height of the image formed on the film. **(6 marks)**
4. (a) Why is it not practical to make a barometer with water instead of mercury? **(4 marks)**
- (b) A force of 200 N and 20 cm long spanner and 150 N and 30 cm long spanner were separately used to loosen the left and right front car tire nuts, respectively. Show that the 30 cm long spanner will be more effective than the 20 cm long spanner. **(5 marks)**
5. (a) Why the recoil velocity of a gun is much less than the velocity of the bullet? Explain by using the principle of conservation of linear momentum. **(4 marks)**
- (b) A hydraulic press consists of a pump, load and two cylinders in which the larger cylinder is eight times the diameter of a smaller one. Use a well labelled cross-sectional diagram of a hydraulic press to determine the value of mechanical advantage of this machine if it is working with efficiency of 90%. **(5 marks)**
6. (a) Carefully study Figure 1. Explain what will happen to the bulbs P and Q in relation to the water levels X and Y when the heater is turned on. **(4 marks)**

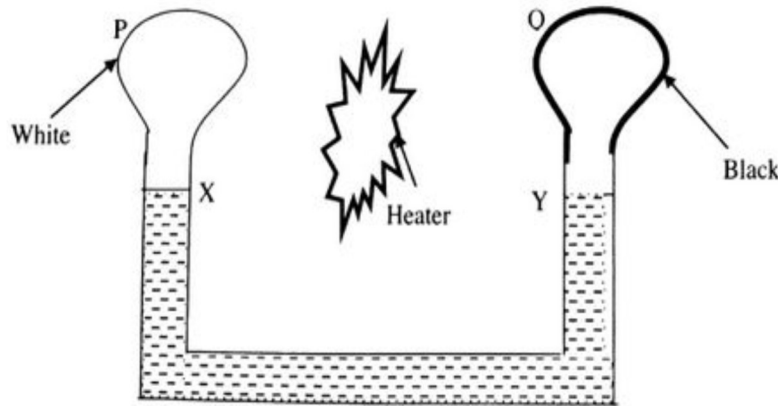


Figure 1

(b) How would the dimensions of a newly constructed Tanzania standard gauge rail line change during the summer and winter days? **(5 marks)**

7. (a) A closed pipe which is opened at one end has a fundamental frequency of 400 Hz. Draw its mode of vibration of air column at its fundamental mode and first overtone. **(5 marks)**

(b) Briefly explain, why does an object appear coloured when light falls onto it? **(4 marks)**

8. (a) Explain why the freezing compartment of a refrigerator is located near the top of the cabinet **(4 marks)**

(b) A spring balance has a scale pan of 0.4N weight attached to it. A load of 2N placed on the pan extends the spring 24mm from its original unstretched length. Calculate the load on the scale pan when the extension is 16mm? **(5marks)**

SECTION C (30MARKS)

Answer **two (2)** questions from this section

9. (a) By drawing circuit diagrams supported by the graphs, compare the charging and discharging processes of a capacitor given that you are provided with a cell (E), a voltmeter (V), a switch (S), a capacitor (C), a resistor (R) and connecting wires. **(6 marks)**

(b) Two wires A and B of the same material and length have the cross-sectional area in the ratio 2:1. If the same potential difference is applied across each wire, comment on the amount of current flowing between the wires A and B. **(6 marks)**

(c) Why a low voltage supply should have a low internal resistance? Explain by using an appropriate formula. **(3 marks)**

10. (a) Explain why the sound produced in a hall with many people is heard clearly than when the hall has few people? **(3 marks)**

(b) A sonometer consists of a taut steel wire fixed between two bridges. Derive the formula for the first, second and third overtones **(9 marks)**

(c) Explain why television or computer screens often become very dusty as compared to their neighbouring surfaces **(3 marks)**

11. (a) The shortest length of the air column in a resonance tube with one end closed and the other end open which resonates to a note of frequency 500 Hz is found to be 160 mm. Calculate the shortest length of the column of air which resonates in similar conditions to a note of frequency 800 Hz. **(5 marks)**

(b) Show how Lenz's law is a special case of the law of conservation of energy. **(3 marks)**

(c) Figure 2 illustrates an electric bell operated by a battery.

(i) What will happen to the armature A when the switch is closed? **(3.5 marks)**

(ii) Describe the function of steel strip S. **(3.5 marks)**

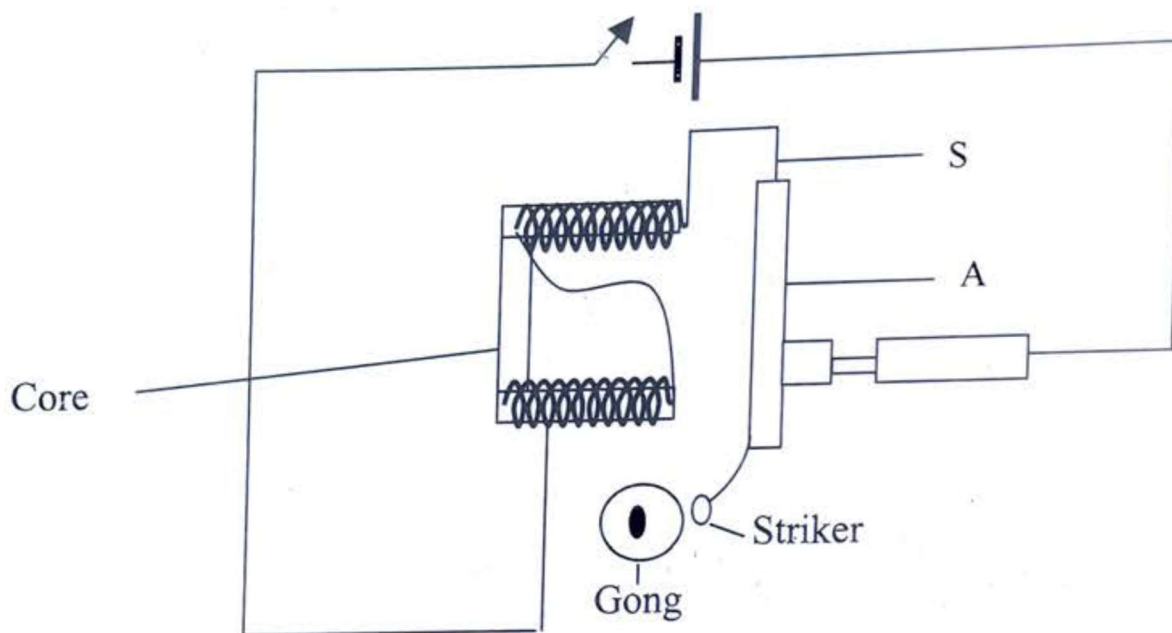


Figure 2